

MyChanges: Tools for the co-designing of housing transformations

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ABSTRACT

MyChanges is a prototype tool for generating and visualizing architectonic modifications of existing housing in co-design projects with inhabitants. Our hypothesis is that giving inhabitants design solutions that would fit individual needs and aspirations, will increase their satisfaction with their house. To arrive at architectonically responsible house transformations, we used a shape grammar system for defining the possible modifications [1]. For empowering inhabitants to understand and explore these modifications to their housing and increase the potential of their participation [2], we developed a mockup tool that comprehends two main parts: shape generation and visualization. The shape generation component is currently a mockup simulation that reproduces some of the generation possibilities of the grammar. For visualizing the outcomes, we developed three different possibilities: i) a semi-immersive visualization where the user utilizes a smart phone to see a 360° render of the site; ii) a fully immersive visualization developed with Unity in which the user with a Head Mounted Display, can freely navigate through the final design; iii) a non-immersive screen-based visualization, where the user, with a tablet device, visualizes a static image of the final design. Interviews and tests with real inhabitants (n=12) were performed to assess user's response to the potential of the tools and preliminary conclusions show that a tool like MyChanges would have acceptance among inhabitants [3].

VIDEO

The video can be found here <https://youtu.be/iWtqOEc4wLc>

KEYWORDS: MyChanges; Generative Design; Virtual Reality; User Interface Design; Augmented Reality.

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